



Susceptibility pattern of methicillin resistance *Staphylococcus aureus* (MRSA) by flow cytometry analysis and characterization of novel lead drug molecule from *Streptomyces* species

Ganesan Govindarajan^{a,c,*}, Priya Mullick^{a,1}, Babu Arulmani Samuel Raj^b, Pachaiyappan Saravana Kumar^c, Mysoon M. Al-Ansari^d, Soundharrajan Ilavenil^e, Robinson David Jebakumar Solomon^a

^a Department of Molecular Microbiology, School of Biotechnology, Madurai Kamaraj University, Madurai, 625021, Tamil Nadu, India

^b School of Environmental Science and Engineering, Guangzhou University, Guangzhou 510006, China

^c CAS Key Laboratory of Tropical Marine Bio-resources and Ecology, Guangdong Key Laboratory of Marine Materia Medica, South China Sea Institute of Oceanology, Chinese Academy of Sciences, 164 Xingang Road West, Guangzhou 510301, China

^d Department of Botany and Microbiology, College of Science, King Saud University, Riyadh 11451, Saudi Arabia

^e Grassland and Forages Division, National Institute of Animal Science, Rural Development Administration, Cheonan 31000, Republic of Korea

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ABSTRACT

Background: Actinomycetes particularly, *Streptomyces* species are producing wide variety of natural products with potential bioactivities. The microbial-derived metabolites hold a strong position to combat emerging and re-emerging antimicrobial drug-resistant pathogens.

Objectives: A diverse group of actinomycetes strains were isolated from unexplored regions of mangrove sediment. Further, a polyphasic approach based on 16S rRNA gene sequence analysis and to evaluate their antibacterial potential against a panel of bacterial pathogens and methicillin resistance *Staphylococcus aureus* (MRSA).

Methods: The mangrove sediment samples were serially diluted with sterile water and plated on inorganic starch agar medium. A total of 20 isolates were pure cultured and 16S rRNA gene sequences were deposited in the public nucleotide databases (GenBank, NCBI). All the isolates were screened for the antibacterial activity by agar overlay method. Further, the susceptibility pattern of MRSA by flow cytometry and fluorescence microscopy was analysed.

Results: These twenty different isolates were grouped under nine major clad and they shared 95–99% sequence identity to the 16S rRNA gene sequences of the genus *Streptomyces* in the public nucleotide databases. Among these strains, the isolates namely JRG-02, JRG-03, JRG-04, JRG-10 and JRG-12 exhibited a broad-spectrum antibacterial activity against Methicillin-resistant *Staphylococcus aureus* (MRSA) and Gram negative bacteria *Klebsiella pneumoniae* MTCC109. Furthermore, we have characterized the antibacterial compound production and its properties from the isolate JRG-02, a potential drug candidate. The culture conditions and various nutrient components of strain *Streptomyces* sp. JRG-02 were optimized for enhanced antibiotics production of the isolate. The FT-IR and LC–MS spectrum analysis envisaged the chemical nature of the substance. The effect of antibacterial compound on the viability of MRSA was alone examined by flow cytometry (FACS) and fluorescence microscopy analysis.

Conclusions: The present study clearly shows that the survival of diverse inhabitants of *Streptomyces* in the mangrove sediments. Hence, the mangrove sediment inhabiting strain *Streptomyces* sp. JRG-02 has potential pharmaceutical activity and genetic diversity.

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* Corresponding author at: Department of Molecular Microbiology, School of Biotechnology, Madurai Kamaraj University, Madurai, Tamil Nadu 625021, India.

E-mail address: govind2610@gmail.com (G. Govindarajan).

¹ These authors have contributed equally to this work.

Introduction

The mangrove ecosystems are a highly untapped source for actinomycetes with the respect to produce pharmaceutically active secondary metabolites and believed as most economically and biotechnologically valuable prokaryotes [1]. The genus *Streptomyces* has commercial importance owing to their unique capacity to synthesize novel secondary metabolites with the wide range of biological activities. They produce abundant, readily dispersed spores which facilitate the bacteria and are well adapted to different habitats including marine environments [2,3]. The screenings of secondary metabolites from mangrove actinomycetes are currently a suitable hotspot, because of richest nutrients, selection pressure and abundance of microbes. The recent studies have been reported that discovery of novel compounds having unique role and chemical backbone from mangrove *Streptomyces* [4,5]. Undoubtedly, mangrove might be developed as a precious source for discovering novel actinomycetes; therefore new chemo-diversity also occurs with salinity act as a driver of bacterial diversity [6]. The new aromatic polyketide metabolite exerted broad spectrum antibacterial effect against different pathogenic microorganisms was identified from mangrove sediment [7]. So far unexplored sources of actinomycetes may have numerous new gene clusters synthesizing novel secondary metabolites for the current drug screening sectors [8]. Similarly, Strong antimicrobial potential actinomycetes were isolated from unexplored Sundarbans mangrove region [9]. The unexploited microbial biodiversity of the mangrove sediment is a potential resource for exploring the novel bioactive compounds. Thus, far only a few studies have been reported for the 16S rRNA based molecular diversity and antibacterial activities of *Streptomyces* sp., from unexplored regions of Karangadu mangrove sediment. Consequently, huge possibility to identify novel actinobacteria and screen them to discover valuable antibacterial compounds against multidrug resistance clinical pathogens. Therefore, the present study investigates the molecular chemotaxonomic position from soil sediments samples collected from Karangadu mangrove forests, Tamil Nadu, India. The study also aims at optimization of medium components and growth conditions for the large-scale production of antibacterial substance and bacterial susceptibility testing of MRSA by live-dead and flow cytometry analyses of the active isolate JRG-02.

Materials and methods

Sample collection and isolation of actinomycetes

The soil sediments were collected from the unexplored regions of mangrove, Latitude: 9°36'N; Longitude: 78°83'E. The sediment samples were collected near the root zone of the mangroves at 0.7–10 cm depth and transferred to the laboratory in aseptic containers for further use. The sediment samples were serially diluted with sterile water and plated on inorganic starch agar medium containing soluble starch - 20 g, NaCl - 0.5 g, KNO₃ - 1.0 g, K₂HPO₄ - 0.5 g, MgSO₄·7H₂O - 0.5 g, FeSO₄ - 20 µM, and agar - 18 g [10]. The isolation medium was supplemented with 75 µg mL⁻¹ of cyclohexamide and 15 µg mL⁻¹ of nalidixic acid to inhibit the fungal and Gram-negative bacterial growth and the plates were incubated at 30 °C for 7–10 days. The bacterial isolates were purely cultured by repeated streak plate technique and stored at 15% glycerol stock for further characterization. All these isolates were screened for the antibacterial activity by agar overlay method [11,12].

Identification of actinomycetes

Twenty different actinomycetes were identified primarily according to the classical morphological criteria, including shape and size of colonies on the plates, substrate mycelium, aerial hyphae view, spore morphology and soluble pigment production [13]. The isolates were selected for further characterization in terms of morphological, physiological, and biochemical parameters. The biochemical tests and extracellular enzyme production tests such as starch, casein, lipid, gelatin liquefaction, catalase, were carried out for the isolates [14]. Similarly, intracellular enzyme assays, nitrate reduction, citrate utilization, triple sugar iron, MR-VP tests were also performed. The physiological studies such as salinity tolerance (NaCl), pH and temperature were determined for all the isolates. Further, molecular analysis was also carried out by using 16S rRNA gene sequencing.

Genomic DNA isolation, 16S rRNA gene amplification

Genomic DNA was isolated for all the isolates as described previously [15] with minor modifications. The 16S rRNA gene was amplified for all the isolates using universal primer 27F (5'-AGA GTT TGA TCC TGG CTC AG - 3') and 1492R (5'-TAC GGC TAC CTT GTT ACG ACTT - 3') [16]. The PCR amplification was carried out with 50 µL reaction mixture consist of reaction buffer (10 mM Tris-HCl pH-8.3, 50 mM KCl, 1.5 mM MgCl₂), 200 µM dNTPs, 0.05 U Taq DNA polymerase enzyme (Sigma, USA), 0.5 µM of each primer and 20 ng template DNA. The thermal cycling conditions were as follows: 5 min at 94 °C for initial denaturation; 31 cycles of 30 s at 95 °C, 1 min 30 s at 54 °C, 2 min at 72 °C; and final extension for 5 min at 72 °C. The amplification reaction was achieved with a thermal cycler (MyCycler, Bio-Rad, USA) and the PCR amplicons were examined by electrophoresis in 1% (w/v) agarose gels to confirm the expected size of the product.

Sequencing and phylogenetic analysis of 16S rRNA gene

The PCR products of 20 actinomycete isolates were purified by QIA quick PCR purification kit (Qiagen, USA). The Purified products were sequenced by Big Dye Terminator v3.1 Cycle sequencing kit (Applied Biosystems, USA) with same the primer set as used in PCR reaction. All the nucleotide sequences were analysed and edited by Bio Edit software [17]. The 16S rRNA gene sequences were analysed by the Blastn algorithm from GenBank database. 16S rRNA gene sequences of closely related type strain sequences were retrieved from RDP database [18]. The Phylogenetic tree has been constructed by the neighbour-joining method using PHYLIP version 3.68 software package and maximum-parsimony methods were used to generate bootstrap values over 1000 re-samplings [19]. The 16S rRNA gene sequences of all the 20 actinomycete isolates were deposited in the GenBank database under accession numbers JX486678 and KU500590 to KU500608.

Antibacterial activity by soft agar overlay method

All the 20 isolates were grown on the inorganic starch agar plate (ISP-4 medium) consists of (g/L): Soluble starch 10; Peptone 1; K₂HPO₄ 1; MgSO₄·7H₂O 1; (NH₄)₂SO₄ 2; NaCl 1; CaCO₃ 2; Agar 2; Trace salt solution 1.0 mL; pH -7.2 - 7.5 and distilled water 1.0 L for 3 days at 28 °C. After 3 days of incubation, the plates were overlaid with log phase cultures of test organisms such as *Escherichia coli* ATCC 3739, *Klebsiella pneumoniae* MTCC 109, *Proteus vulgaris* MTCC 426, *Pseudomonas aeruginosa* ATCC 10145, *Bacillus subtilis* MTCC 441, *Staphylococcus aureus* ATCC 6518, *Salmonella typhi* MTCC 350733, *Staphylococcus aureus* MTCC 3160 and Methicillin-resistant *Staphylococcus aureus* (MRSA) (clinical

isolate obtained from a speciality hospital), in 0.5% soft agar nutrient medium. Plates were incubated at 37 °C for 12–24 h [20]. The microorganisms were purchased from American Type Culture Collection (ATCC, Manassas, VA, USA), and Microbial Type Culture Collection (MTCC, IMTECH, Chandigarh, India). MRSA is a clinical isolate which is obtained from the Madurai Kamaraj University Hospital, India.

Optimization of media and culture conditions

Among these isolates, the isolate JRG-02 was selected for further antimicrobial production and its biological action. The effect of carbon, nitrogen source, NaCl concentration, various pH and temperature were examined for the growth and antibacterial compound produced by the *Streptomyces* sp. JRG-02. For this study, nutrients were amended with the basal medium containing Na₂HPO₄ 2.9 g, KH₂PO₄ 2.3 g, NH₄Cl 1.0 g, MgSO₄·7H₂O 0.5 g, FeSO₄·0.002, CaCO₃ 0.5, and trace element salt solution of 5 mL from stock solution (ZnSO₄·xH₂O 100 mg, H₃BO₃ 300 mg, CoCl₂·xH₂O 200 mg, MnCl₂·xH₂O 30 mg, Na₂MoO₄·xH₂O 30 mg, NiCl₂·xH₂O 20 mg, CuCl₂·xH₂O 10 mg and distilled water 1.0 L) [21]. The classical media optimization was performed by a one factor at a time (OFAT) approach in which one factor is changed and other factors kept as constant. In order to select the suitable carbon source, 1% of glucose, sucrose, maltose, fructose, starch and glycerol were used independently and supplemented with basal medium. For the nitrogen source optimization, organic nitrogen sources such as yeast extract, beef extract, peptone, tryptone, and urea were given to the minimal medium for maximum antimicrobial compound production. Likewise, the effect of varying pH ranges (pH 6.0 to pH 12) and temperature between (25 °C–45 °C) was determined for the growth and antibacterial compound production. The optimum NaCl concentration was studied by varying concentration from 0.5% to 3% for antibacterial compound production. The effectiveness of suitable parameters was selected based on their antibacterial activity against test organism methicillin resistance *Staphylococcus aureus* (MRSA) on Muller Hinton agar media (MHA). All these experiments were performed in triplicates. The outcome of the OFAT method was used for the large-scale antibacterial compound production.

Large scale production, extraction and purification of bioactive compound

Seed culture was prepared by inoculating single colony of pure culture into 250 mL Erlenmeyer flask containing 50 mL of production medium at 28 °C for 36 h. 1% of seed culture was inoculated into fermenter (New Brunswick Scientific model number Bioflow/Celligen 1150 from Eppendorf, NJ, USA) containing 4000 mL of sterile production medium. Desired pH, temperature and agitation speed 120 rpm were programmed in the control unit. Supply of acid and base were ensured to maintain the optimum pH. The production medium consisted of (g/L): Maltose - 10 g, Tryptone - 5 g, Na₂HPO₄ - 2.9 g, KH₂PO₄ - 2.3 g, NH₄Cl - 1 g, MgSO₄·xH₂O - 0.5 g, FeCl₃ - 0.002 g, CaCO₃ - 0.5 g and trace element salt solution 5.0 mL.

After 7 days of incubation, the culture broth was harvested by using harvesting tube and the cell free supernatant was separated by centrifugation at 10,000 rpm for 30 min. Extraction was done twice by adding equal volume of chloroform to the supernatant and concentrated by rotational vacuum concentrator RVC2-18CD (CHRIST, Germany). Finally, the bioactive compound residue was dissolved in chloroform for further use. Purification of the bioactive compound was done by silica column chromatography followed by gradient solvent system (methanol:chloroform 30:70). Subsequently, thin layer chromatography (chloroform: methanol = 7:3) was adopted for those fraction validations. The antibacterial activ-

ity of the fractions was tested against pathogens by disc diffusion method [22] and the purified fraction was collected by high performance liquid chromatography [7,23] in reverse phase silica column with shim-pack CLC ODS (4.6 mm × 15 mm), UV absorption at 280 nm with flow rate 1.0 mL/min.

Spectral analysis of antimicrobial compound

FT-IR spectral analysis of the purified active fractions was examined by SHIMADZU FT/IR-8400S spectrophotometer equipped with MCT detector and germanium-coated KBr plate beam splitter. The electrospray ionization mass spectrometry (ESI-MS) mass spectra of active compound JGP1 was analysed with a liquid chromatography-ion trap mass spectrometer (LCQ Fleet, Thermo Fisher Instruments Limited, USA) at the negative ion detection mode. The samples were introduced into the ion source by the infusion method at the flow rate 1.0 mL/min. The capillary voltage of the mass spectrometer was 33 V, with source voltage 4.98 kV for the mass scale (*m/z* 200–1000).

Minimum inhibitory concentration of the isolated compound JGP1

Minimal inhibitory concentration (MIC) of the active compound was evaluated against test organisms (NCCLS) by broth dilution method as per Clinical and Laboratory Standards Institute (CLSI) in triplicates.

Antibacterial susceptibility test by flow cytometry and fluorescence microscopy

Log-phase culture of MRSA cells were (OD₆₀₀ = 0.5) treated with purified antibacterial compound for 3 h at 37 °C. Three different minimal inhibitory concentration as followed (1 × MIC, 3 × MIC and 7 × MIC) and dimethyl sulfoxide (DMSO) is used as a solvent control. Subsequently the untreated and treated cells were collected by centrifugation at 5000 rpm for 10 min and washed twice with PBS [7]. After washing, the cells were stained with 10 µL of acridine orange (AO: Sigma Aldrich) from (10.0 mg/mL stock) and incubated at dark for 30 min. Subsequently, the cells were stained with 1 µL of propidium iodide (PI: Sigma Aldrich) from (1.0 mg/mL stock) and kept in dark condition for 5 min. 10 µL stained cells were transferred to the microscopic slide aseptically and covered with a cover slip. The cells are visualized under fluorescence microscope with a filter range of 480 nm and 567 nm, 20× magnifications. In the flow cytometry analysis, the above-mentioned concentration of antimicrobial compound was used for the treatment of log phase culture of MRSA cells for 3 h at 37 °C. After washing with PBS buffer, the cells were stained with propidium iodide (PI 1.0 mg/mL stock), and kept in dark condition for 5 min. PI-stained cell aliquots were subjected for MRSA cell membrane damage by using flow cytometry on FACS Calibur (BD Biosciences, Oxford, UK) and the results were analysed with cell Quest Pro software (BD Biosciences, Oxford, UK) [24]. The experiment was performed in triplicates.

Results

Isolation of actinomycetes

Twenty different bacterial isolates were identified as actinomycetes based on their colony morphological features such as musty odour, sporulation, dimorphic mycelial forms as aerial and substrate mycelium. The actinomycetes which were exhibited in different colony morphology characteristics isolates from mangrove sediments were shown in Fig. 1. After seven days of incubation, they were showing the distinct difference in their aerial and substrate view and it also produced soluble pigments like brown,

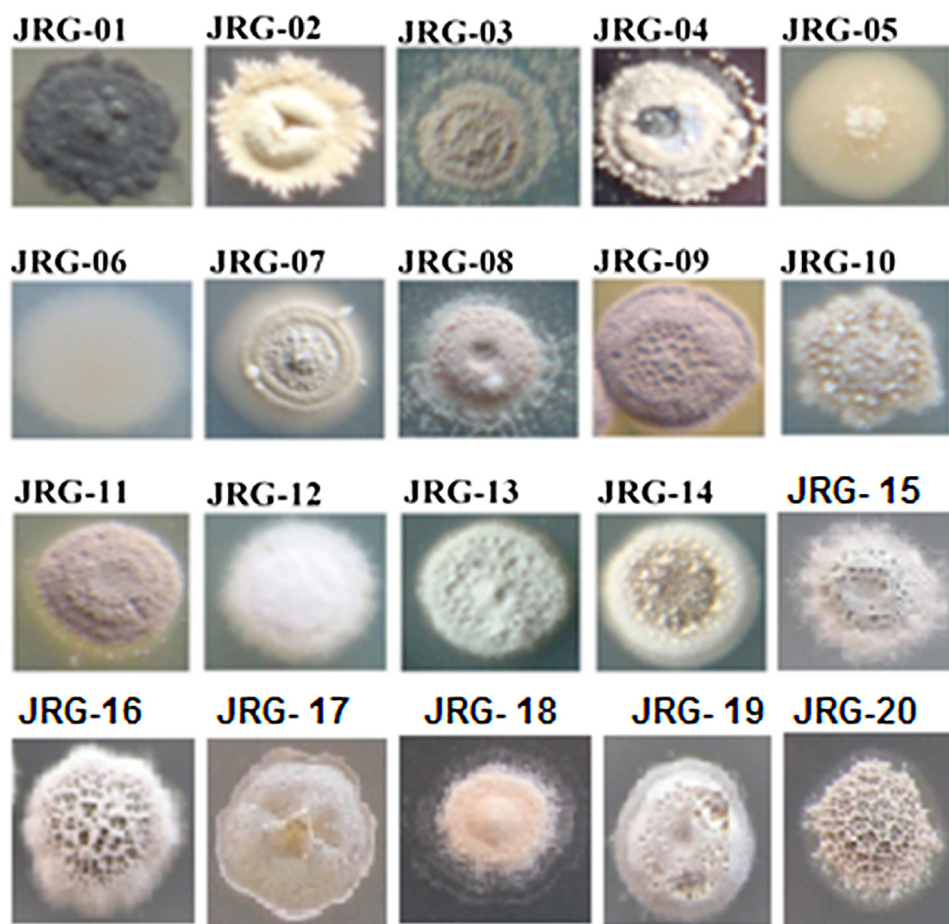


Fig. 1. Aerial view of seven days-old culture of actinomycetes from mangrove sediments.

yellow, black, grey and orange. Some of them produce white spores with exudates of fine watery droplets on their surface.

Physiological and growth characteristics of actinomycetes

The physiological and biochemical characteristic features of all the actinomycetes isolates were examined and the results were summarized. The Gram-positive, spore forming and non-motile nature of the isolates revealed that the isolates were belongs to the largest antibiotic producing genus *Streptomyces*. All the isolates were grown well in presence of sodium chloride up to (w/v) 7.0%. These strains grew well at pH 6 to 8. These strains grown up to 37 °C and the optimum growth temperature were found at 28 °C–30 °C. All the isolates were displayed a positive role for amylase and extracellular enzyme production. The isolates JRG-02, JRG-03, JRG-05, JRG-06, JRG-07, JRG-08, JRG-11, JRG-12, JRG-13, and JRG-15 could utilize cellulose as a carbon source. The isolate JRG-01, JRG-05, JRG-06, JRG-07, JRG-08, JRG-09, JRG-10, JRG-11, JRG-12, JRG-15, JRG-16, and JRG-20 were displayed positive action in casein hydrolysis, which denotes an extracellular casein enzyme production. All these isolates produced extracellular the catalyse enzymes except the isolate JRG-01, JRG-03, JRG-13, and JRG-18.

PCR amplification and phylogenetic diversity of actinomycetes

The genomic DNA of all the 20 isolates were used as a template and 16S rRNA gene was amplified from each isolate by using universal primer sets such as 27F and 1492R specific primers. All the

20 isolates showed good amplification of 16S rRNA gene of 1500 bp length. The 16S rRNA gene sequence analyses were carried out to identify the taxonomic position of the twenty isolates. The phylogenetic analysis showed that the isolates were the members of the GC rich *Streptomyces*. They shared 95–99% sequence identity to the 16S rRNA gene sequences of the genus *Streptomyces* in the public nucleotide databases (GenBank, NCBI). The phylogenetic trees revealed inter and intra specific polymorphism which is grouped under nine major clusters (Fig. 2). Based on the 16S rRNA phylogenetic analysis, the cluster I consist of the isolate JRG-05 and isolate JRG-17, they are closely related to the *S. chrestomyceticus* (98.1%) and *S. sioyaensis* (99.2%), respectively. The cluster II had the isolates JRG-20 and JRG-18 in the same clade. These isolates were the very close relative of a type strain the *S. varigatus* that shared 98.3% sequence similarity. In cluster III the isolates JRG-06, JRG-13 and JRG-02 were similar to the *S. canarius*, *S. chartreusis* and *S. laculatispora* with sequence similarity of 98.4%, 98.6% and 98%, respectively. The cluster IV contained the three isolates such as JRG-14, JRG-08, and JRG-01 which were highly related to the *S. katrae* (98%), *S. albolongus* (99.8%) and *S. globosus*, respectively. Cluster V contained two isolates namely JRG-11 and JRG-07 they were closely related to the type strain *S. coeruleoprurus* (98.7%) and *S. thermoviolaceus* (98.4%). The cluster VI and VII consist of isolate JRG-19 and JRG-12 these share 95.9% and 99.9% sequence similarity with *S. chromofuscus* and *S. rubrogriseus*, respectively. The three isolates (JRG-04, JRG-16 and JRG-15) were grouped in the cluster VIII. These isolates were closely related to the nearest type strain of *S. longisporuber*, *S. speibonae* and *S. viridodiastaticus* with the sequence similarity of 99.6%, 99.6% and 99.4%, respec-

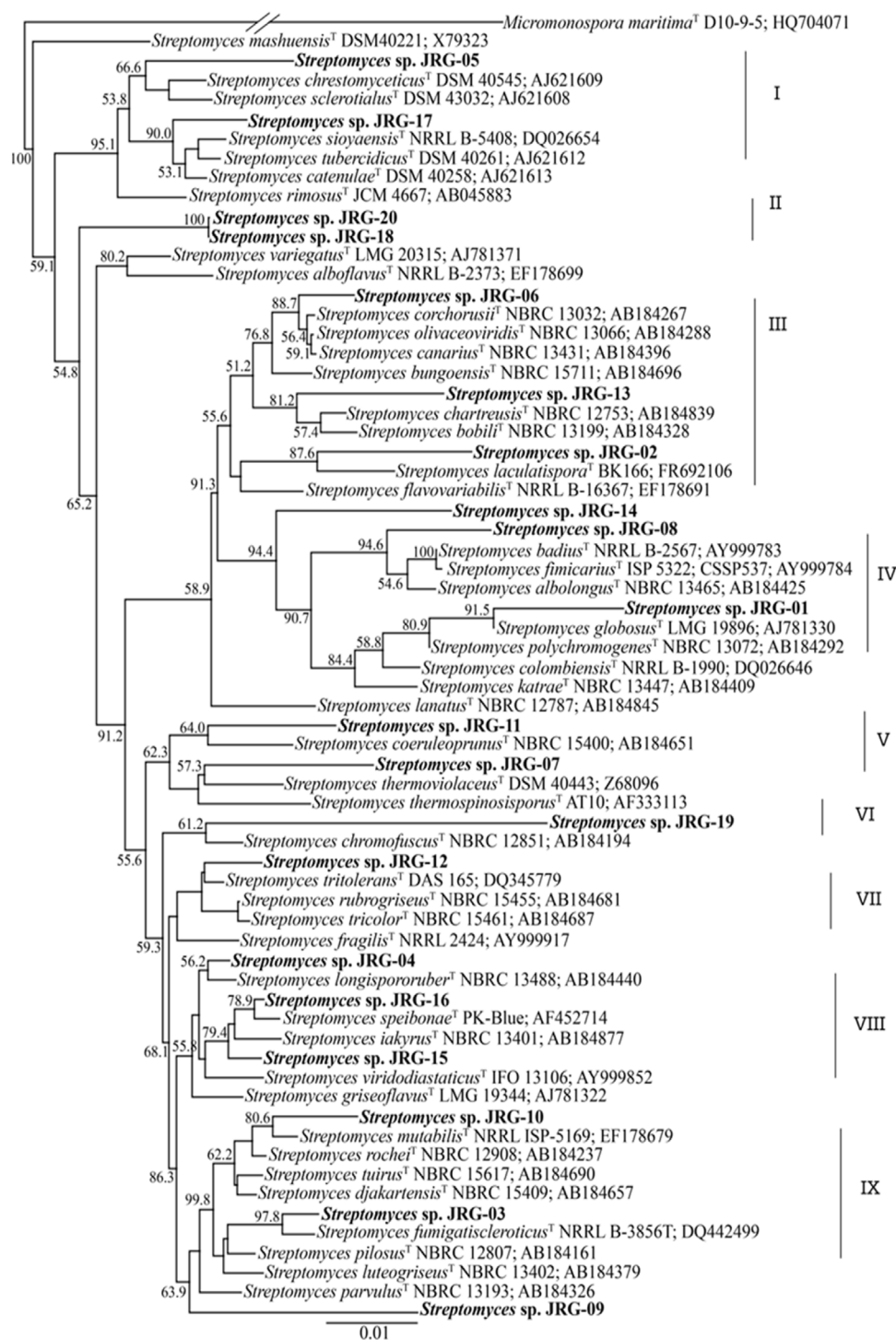


Fig. 2. Neighbor-joining phylogenetic dendrogram illustrating the clustering of partial 16S rRNA gene sequences of the *Streptomyces* isolates JRG-01 to JRS-20. The tree topology was analysed by neighbour-joining and parsimony analysis with bootstrapping of 1000 replicates using the PHYLIP inference package. Bootstrap values >50% was illustrated adjacent to each node. The Roman number I to IX represents the clusters of the *Streptomyces* isolates, and *Micromonospora maritima* D10-9-5 (HQ704071) is the out group. The scale bar (0.01) indicates the nucleotide substitution level.

tively. The cluster IX had three isolates, namely JRG-10, JRG-03 and JRG-09. The isolate JRG-10 exhibited 99.4% sequence similarity with *S. rochei*. The isolate JRG-03 showed 99.3% sequence similarity with the type strain of *S. fumigatiscleroticus*. The isolate JRG-09 was closely related to the *S. djakartensis* with 98.1% sequence similarity. The closest relative of these 20 isolates and their sequence identity were shown in Table 1. These 20 distinctive isolates were named as *Streptomyces* sp. JRG-01 to JRG-20 on the basis of their

detachment from close relatives with nearest sequence identity value.

Screening of antibacterial substance producers

The isolates namely JRG-02, JRG-03, JRG-04, JRG-10 and JRG-12 exhibited a broad spectrum antibacterial activity against test organisms including MRSA (Table 2). In the clade I and II con-

Table 1
16S rRNA gene sequence similarity of *Streptomyces* isolates.

S.No.	Isolate	Closely related type strain	Sequence similarity
1	<i>Streptomyces</i> sp. JRG-01	<i>S. globosus</i> ^T LMG 19896; AJ781330	99.3 %
2	<i>Streptomyces</i> sp. JRG-02	<i>S. laculatispora</i> ^T BK166; FR692106	98.0%
3	<i>Streptomyces</i> sp. JRG-03	<i>S. fumigatiscleroticus</i> ^T NRRL B-3856T; DQ442499	99.3%
4	<i>Streptomyces</i> sp. JRG-04	<i>S. longisorporuber</i> ^T NBRC 13488; AB184440	99.6%
5	<i>Streptomyces</i> sp. JRG-05	<i>S. chrestomyceticus</i> ^T DSM 40545; AJ621609	98.1%
6	<i>Streptomyces</i> sp. JRG-06	<i>S. canarius</i> ^T NBRC 13431; AB184396	99.3%
7	<i>Streptomyces</i> sp. JRG-07	<i>S. thermoviolaceus</i> ^T DSM 40443 Z68096	98.4%
8	<i>Streptomyces</i> sp. JRG-08	<i>S. albolongus</i> ^T NBRC 13465; AB184425	99.8%
9	<i>Streptomyces</i> sp. JRG-09	<i>S. djakartensis</i> ^T NBRC 15409; AB184657	98.1%
10	<i>Streptomyces</i> sp. JRG-10	<i>S. rochei</i> ^T NBRC 12908; AB184237	99.4%
11	<i>Streptomyces</i> sp. JRG-11	<i>S. coeruleoprunus</i> ^T NBRC 15400; AB184651	98.7%
12	<i>Streptomyces</i> sp. JRG-12	<i>S. rubrogriseus</i> ^T NBRC 15455; AB184681	99.9%
13	<i>Streptomyces</i> sp. JRG-13	<i>S. chartreusis</i> ^T NBRC 12753; AB184839	98.6%
14	<i>Streptomyces</i> sp. JRG-14	<i>S. katrae</i> ^T NBRC 13447; AB184409	98.0%
15	<i>Streptomyces</i> sp. JRG-15	<i>S. viridodiataticus</i> ^T CSSP719; AY999852	99.4%
16	<i>Streptomyces</i> sp. JRG-16	<i>S. speibonae</i> ^T PK-Blue; AF452714	99.6%
17	<i>Streptomyces</i> sp. JRG-17	<i>S. siayaensis</i> NRRL B-5408; DQ026654	99.2%
18	<i>Streptomyces</i> sp. JRG-18	<i>S. variegatus</i> ^T LMG 20315; AJ781371	98.3%
19	<i>Streptomyces</i> sp. JRG-19	<i>S. chromofuscus</i> ^T NBRC 12851; AB184194	95.9%
20	<i>Streptomyces</i> sp. JRG-20	<i>S. variegatus</i> ^T LMG 20315; AJ781371	98.3%

Table 2
Antibacterial activity of lineages of *Streptomyces* in clusters I to IV.

Actinomycetes isolates Bacterial pathogens	JRG-02 Zone of inhibition (mm)	JRG-03	JRG-04	JRG-10	JRG-12
MRSA	33 ± 0.20	20.00	55 ± 0.35	33 ± 0.20	32 ± 0.20
<i>S. aureus</i> MTCC 3160	46 ± 0.09	29 ± 0.05	53 ± 0.26	28 ± 0.05	30.00
<i>S. aureus</i> ATCC 6518	44 ± 0.05	27 ± 0.25	54 ± 0.05	29 ± 0.05	31 ± 0.28
<i>B. subtilis</i> MTCC 441	50 ± 0.05	30.00	52 ± 0.15	50 ± 0.05	44 ± 0.10
<i>E. coli</i> ATCC 3739	55 ± 0.09	35.00	57 ± 0.05	34 ± 0.11	40.00
<i>S. typhi</i> MTCC 733	46 ± 0.05	26 ± 0.11	50 ± 0.11	31 ± 0.09	24 ± 0.11
<i>P. aeruginosa</i> ATCC 10145	50.00	34 ± 0.05	57 ± 0.20	34 ± 0.11	43 ± 0.28
<i>P. vulgaris</i> MTCC 426	34 ± 0.05	33 ± 0.05	41 ± 0.11	35.00	34 ± 0.20
<i>K. pneumoniae</i> MTCC109	40 ± 0.35	20.00	30 ± 0.20	30.00	30.00

The data shown are mean values of three replicates ± standard deviation.

Isolate showed no activity: JRG-01, JRG-05, JRG-06, JRG-07, JRG-08, JRG-09, JRG-11, JRG-13, JRG-14, JRG-15, JRG-16, JRG-17, JRG-18, JRG-19 and JRG-20.

sist of the *Streptomyces* isolates namely JRG-05, JRG-17, JRG-20 and JRG-18, showed no antimicrobial activity against the tested pathogens. The isolate JRG-02 grouped in the cluster III, which showed highest antibacterial action against *S. aureus* ATCC, *S. aureus* MTCC 3160, MRSA, *E. coli* ATCC3739, *P. aeruginosa* ATCC 10145 and *B. subtilis* MTCC 441. It showed moderate antibacterial efficacy towards *P. vulgaris* MTCC 426 and *K. pneumoniae* MTCC 109. The *Streptomyces* isolates JRG-06 and JRG-13 showed no antibacterial activities against the screened test organisms. The actinomycetes isolate namely JRG-14, JRG-08, JRG-01 JRG-11, JRG-07 and JRG-19 was present in the cluster IV, V and VI, respectively. They do not exhibit antibacterial activity against test pathogens. The *Streptomyces* isolate of the cluster VII displayed moderate effect against Gram negative bacteria such as *E. coli* ATCC 3739, *P. aeruginosa* ATCC 10145, *P. vulgaris* MTCC 426 and *K. pneumoniae* MTCC109. Similarly, it displayed intermediate antagonistic activity against Gram-positive bacteria namely *S. aureus* ATCC 6518, *S. aureus* MTCC 3160 and MRSA. The cluster VIII consists of three different *Streptomyces* isolate JRG-04, JRG-16 and JRG-17. Among these, isolate JRG-04 showed highest antibacterial activity against all the tested pathogens. Apart from this, the isolate JRG-10, JRG-03 and JRG-09 can reside in the cluster IX. The isolate JRG-10 and JRG-03 showed moderate antibacterial activity against the tested Gram positive and Gram-negative bacterial pathogens. The isolate JRG-09 do not exhibit antibacterial action against any one of the tested pathogens, even though in the same cluster. The isolate JRG-02 was identified as potential drug candidates for further subsequent studies.

Effect of carbon and nitrogen source on antibacterial compound production

To determine the optimum carbon source for growth and bioactive compound production of the most active isolate *Streptomyces* sp. JRG-02 were tested with different carbon sources. Maltose supplemented medium showed higher antibacterial activity (160 U/mL ±1.00) against *Staphylococcus aureus* and moderate effect (140 U/mL ±0.57) was shown in starch containing medium. Poor antibacterial activity was observed when using glucose, lactose, fructose, sucrose as carbon sources shown in Fig. 3A. The effect of nitrogen source on biomass and bioactive compound production of *Streptomyces* sp. JRG-02 was studied. A wide variety of organic nitrogen source were tested, amongst the tested nitrogen sources, tryptone supplemented to the production medium had maximum biomass and antimicrobial activity (180 U/mL ±0.57) shown in Fig. 3B than other tested nitrogen sources. Peptone and urea grown culture showed moderate growth with poor antibiotic activity (95 U/mL ±0.57).

Effect of NaCl concentration on antimicrobial activity

The effect of salt (NaCl) concentration on growth and antimicrobial compound production of *Streptomyces* sp. JRG-02 was studied by varying salt concentration ranges from 0 to 3% NaCl. The result showed that the maximum growth and antibacterial activity against *S. aureus* (170 U/mL ±0.57) was observed at 1% of NaCl containing basal medium along with optimized nutrients and culture

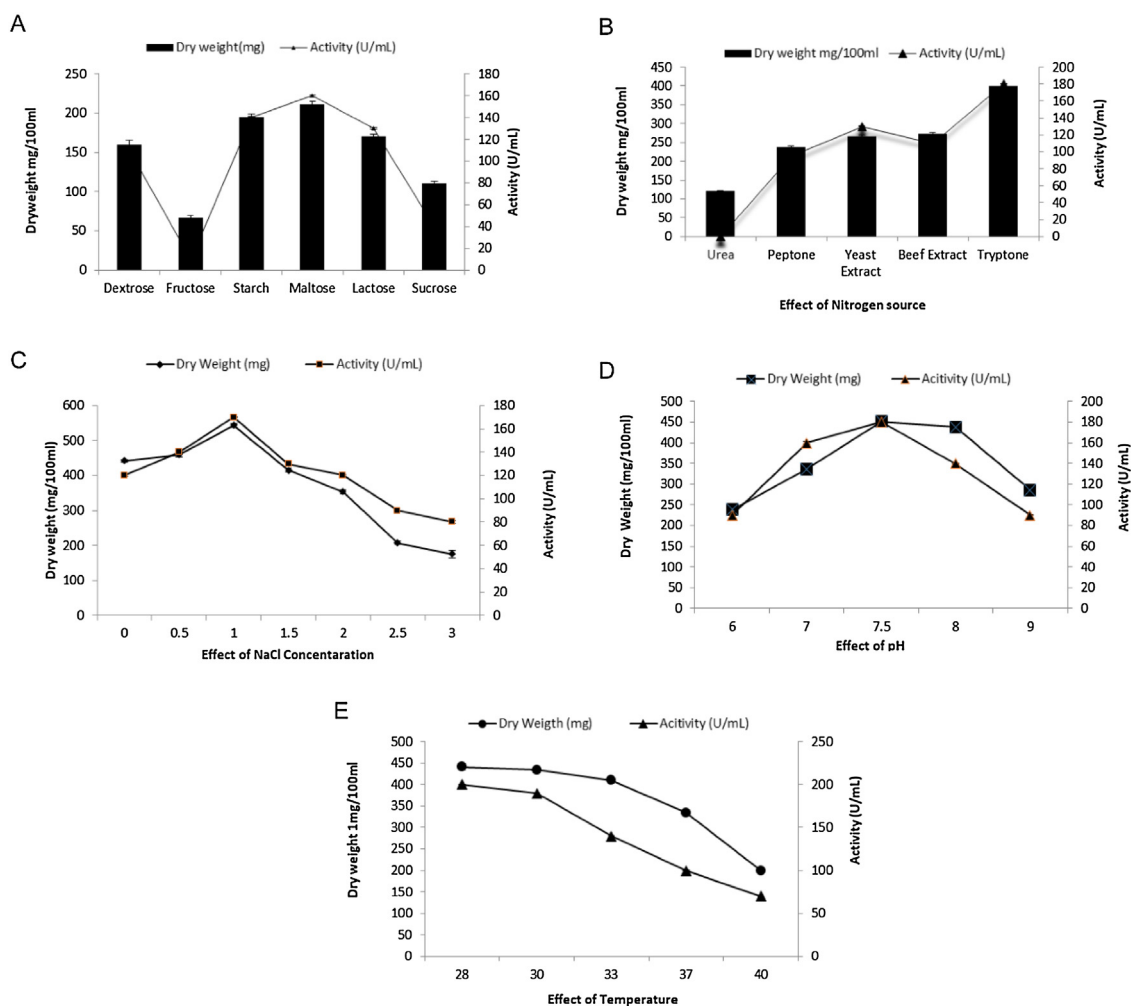


Fig. 3. Effect of (A) carbon source, (B) nitrogen source, (C) NaCl concentration, (D) pH, (E) temperature on growth and antimicrobial compound production by *Streptomyces* sp., JRG-02.

conditions (Fig. 3C). The less antibacterial activity was observed at 3% NaCl and the organism also showed moderate antibacterial activity against *S. aureus* (80 U/mL $\pm$ 1.15) in the absence of NaCl concentration. When increasing salt concentration above 1.5%, it affects the growth and antibacterial activity of the isolate JRG-02. Further increasing the salt concentration declined the biomass and antibacterial compound production of the isolate JRG-02. The optimized salt (NaCl) concentration was used for further optimization studies.

Effect of pH and temperature on growth and antimicrobial activity

Optimum pH for effective growth and antibacterial activity of isolate JRG-02 was observed at pH 7.5 (Fig. 3D). Poor antibacterial activity was observed at below pH 7 and above the neutral pH. Maximum antibacterial activity was found to be neutral compare to acidic and alkaline pH. The effect of temperature ranging from (28 °C to 40 °C) on antibacterial substance production for the strain *Streptomyces* sp. JRG-02 was studied, and the maximum antibacterial activity and growth was found to be 28 °C and moderate antibacterial compound production and growth was observed at 33 °C–37 °C (Fig. 3E). The optimized condition and media components were used for further compound production.

Purification and spectral characterization of antibacterial compound

Production of bioactive compound was carried out by using laboratory scale fermenter. The cell free supernatant was collected and the bioactive compound was extracted with equal volume of chloroform and quantified. The isolate JRG-02 may able to produce 300 mg/L of bioactive compound in the batch fermentation. The extracted bioactive compound was further subjected to thin layer chromatography, column chromatography and also purity was checked by HPLC analysis. The active compound retention time was identified at 1.7 min. The chemical composition of antibacterial compound produced by JRG-02 isolate was preliminarily investigated by FT-IR spectroscopy. The FT-IR spectrum showed the stretching frequencies at 3284.77 cm^{-1} , 1639.49 cm^{-1} and 1658.78 cm^{-1} indicated the presence of hydroxyl and ketone ($\text{C}=\text{O}$ and $\text{CN}=\text{C}$) functional groups in the active compound. Subsequently, ESI-MS analysis of active compound of JGP1 showed the peak at m/z 372.13 ($[\text{M}+\text{H}]^+$) and 393.28.17 ($[\text{M}+\text{Na}]^+$) adduct (Fig. 4).

Minimal inhibitory concentration (MIC) determination

The antimicrobial property of the active compound was determined against various test microorganisms and the results were shown in Table 3. The active fractions showed strong antibacterial activity against both Gram-positive and Gram-negative bacteria.

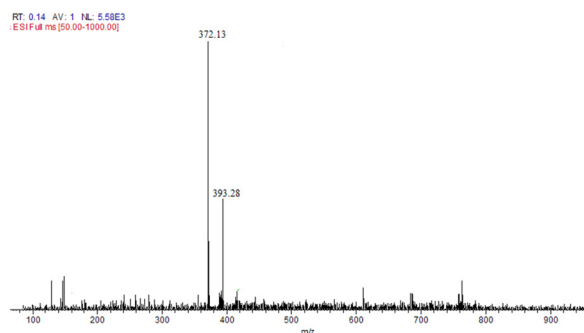


Fig. 4. ESI-MS spectrum of active compound produced from *Streptomyces* sp., JRG-02.

Table 3
Antibacterial activity and MIC determination against various test organisms.

Test organism	Extract Zone of inhibition (mm)*	Compound (JGP1) MIC value ($\mu\text{g}/\text{mL}^{-1}$)
<i>S. aureus</i> MTCC 3160	17 $\pm$ 0.1	2.5–5.0
MRSA	18 $\pm$ 0.1	2.5–5.0
<i>P. aeruginosa</i> ATCC 10145	19 $\pm$ 0.1	0.6–1.25
<i>E. coli</i> ATCC 3739	21 $\pm$ 0.1	0.3–0.6
<i>B. subtilis</i> MTCC 441	21 $\pm$ 0.2	0.3–0.6
<i>K. pneumoniae</i> MTCC109	18 $\pm$ 0.2	2.5–5.0
<i>P. vulgaris</i> MTCC 426	19 $\pm$ 0.1	1.25–2.5

*The values are represented as mean of triplicate experiments.

The minimal inhibitory concentration (MIC) of purified compound has been evaluated and the results indicated that the compound has very strong activity against different test microorganisms including MRSA a clinical pathogen with MIC range of 2.5 μg to 5.0 μg mL^{-1} .

MRSA cell damage assay by flow cytometry and fluorescence microscopy analysis

In addition to MIC, we have also examined MRSA cell membrane damage analysis by using nucleic acid staining dyes such as acridine orange (AO) and propidium iodide (PI). Acridine orange is membrane permeable dye, penetrates to all the dead and live cells and showed in green colour. Propidium iodide is membrane impermeable dye which binds to the dead cells and fluorescence red in colour. Increased number of live cells (green colour) was observed in untreated MRSA cells (Fig. 5a(i), 5a(ii) and 5a(iii)) when compared to the treated cells. In contrast, 14.4% dead cells were observed at $1 \times \text{MIC}$ (7.5 μg mL^{-1}) concentration after bioactive compound (JGP1) exposure for 3 h (Fig. 5b(i), 5b(ii) and 5b(iii)). Antibacterial compound treatment at above MIC concentrations such as ($3 \times \text{MIC}$ 7.5 μg mL^{-1} and $7 \times \text{MIC}$ 17.5 μg mL^{-1}) for 3 h showed increased cell membrane damage followed by cell death of 18% and 79%, Fig. 5c(i), 5c(ii) and 5c(iii) and Fig. 5d(i), 5d(ii) and 5d(iii) respectively. The present fluorescence microscopy report showed that the lethal effect of active compound is mainly in concentration dependent manner (Fig. 6a, 6b and Fig. 6c, 6d).

Discussion

Mangroves are the unique part of coastal bodies in tropical and subtropical regions, high moisture, high salinity and hypoxia tolerant and extremely productive ecosystems [25]. The taxonomic and metabolically diverse *Streptomyces* species were continually isolated from marine habitats [26]. The mangrove bio network is noteworthy from other ecosystems by periodic tidal force and various environmental parameters such as salinity, tidal gradient and nutrient availability which are strong selection pressure for effective

metabolic pathway adaptations. This strong selection can lead to generating unusual and new metabolites [27]. The present study revealed that the *Streptomyces* species were most predominate survivors in the Karangadu mangrove sediments, a poorly studied environment. These isolates were characterized based on their morphological features such as Gram-positive, nonmotile colonies, dimorphic mycelial forms as aerial and substrate mycelium, spore production and musty odour [10]. The good growth pattern of the antibiotic producing isolates in higher salt concentration (7%) with different ranges (pH 6–9) suggested that the extreme environment conditions might provide a great opportunity to the *Streptomyces* sp., for synthesis the antimicrobial substance. The extracellular enzymes and antibiotic substance producing a potential of isolates revealed that the *Streptomyces* isolates had a higher metabolic rate in the mangrove sediments. Previous reports also suggested that the mangrove sediments are a rich source of actinomycetes population for various enzymes and antimicrobial molecules [28]. These results provide that the novel marine actinomycetes genera and species for the discovery of new chemically distinct entity with significant biological activity in the mangrove sediments.

The 16S rRNA genes sequence is molecular clock for the persistence of phylogenetic relationships of the microorganism, appraisal of their diversity, detection and quantification of definite populations in the particular environment [29]. The 16S rRNA based phylogenetic analysis suggested that the isolates derived from mangrove sediments belong to actinomycetes lineages. The maximum parsimony and neighbour joining tree topology revealed that the isolates fall in the *Streptomycineae*, the largest antibiotic producing genus *Streptomyces*. These data clearly indicate the diversity of *Streptomyces* lineages in the mangrove sediment, which is one of the richest metabolically active communities in the mangrove ecosystem. Similarly, metabolically active and biotechnologically important group of *Streptomyces* strains were isolated and characterized from mangrove sediments in the Manakudi estuary [30]. This study shows the importance and occurrence of pharmaceutically valuable *Streptomyces* sp., from different mangrove sediment. The total actinomycetes count and morphology-based biodiversity of actinomycetes were reported earlier in the Karangadu mangrove sediments [31]. Similarly, the diversity of endophytic actinomycetes and its antimicrobial potential from Karangadu mangrove ecosystem was reported [32]. Hence, we have reported the diversity of actinomycetes from poorly researched Karangadu mangrove sediments by using 16S rRNA molecular clock.

In the present study, the *Streptomyces* sp., (JRG-02, JRG-03, JRG-04, JRG-10, and JRG-12) from the different clusters were active against both Gram-positive and Gram-negative human pathogenic bacteria. The identification of genus *Streptomyces* strains in our study and their antibacterial potential against pathogenic microorganisms is an excellent correlation with previously reported data [33]. It was noticed that the genus *Streptomyces* are amongst mangrove actinomycetes of natural product producer. These data reveal that the mangrove sediment derived *Streptomyces* can be an interesting source of novel secondary metabolites. Our studies clearly shows that the survival of diverse inhabitants of *Streptomyces* in the mangrove sediments with potential biological activity and genetic diversity. The 16S rRNA based molecular analysis clearly indicates the occurrence and distribution of *Streptomyces* population in Karangadu mangrove sediment, Tamil Nadu, India. Different *Streptomyces* sp., isolated from mangrove ecosystem had the ability to inhibit the growth of wide variety of pathogenic microorganisms including methicillin resistant *S. aureus* (MRSA). Consequently, the five selected *Streptomyces* sp., may hold a prominent position for our current antibiotic screening programme against MRSA pathogen when compared to other *Streptomyces* species from the unexplored mangrove sediment.

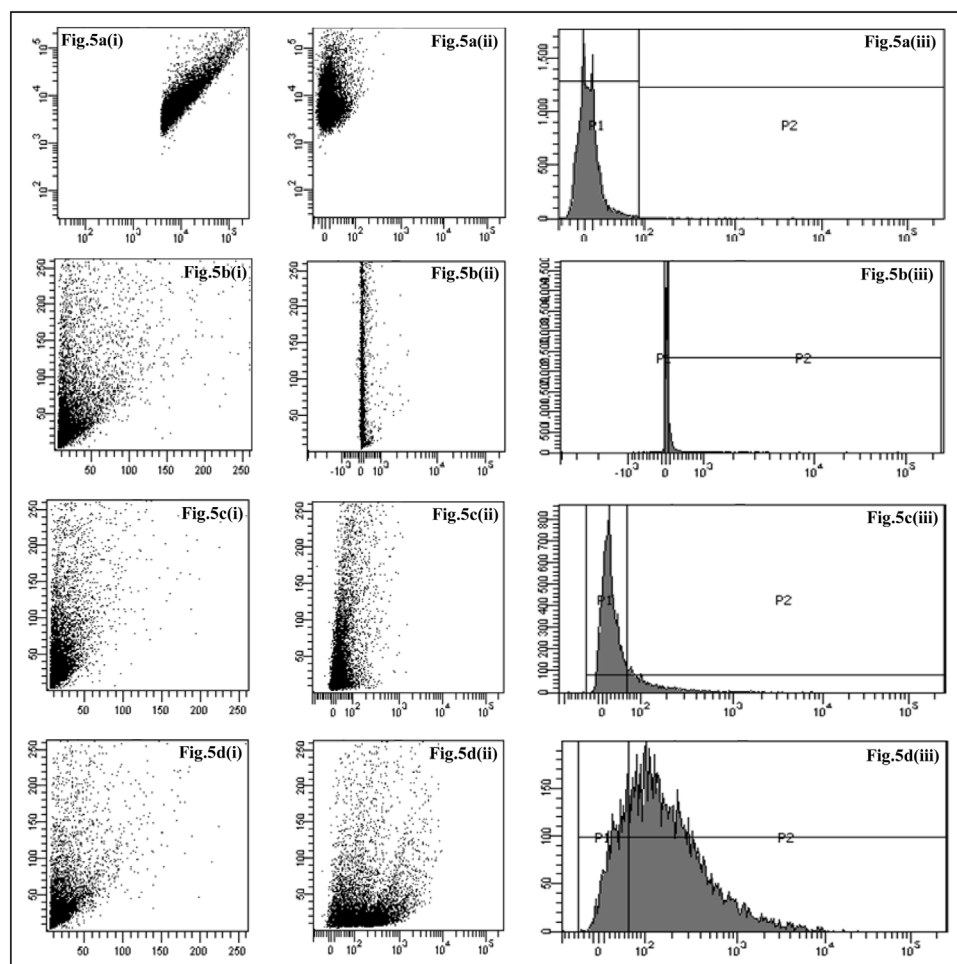


Fig. 5. MRSA cells stained with PI by Flow cytometry analysis. 5a(i), 5a(ii) and 5a(iii) were untreated and 5b(i), 5b(ii) and 5b(iii) were treated with $1 \times$ MIC. 5c(i), 5c(ii) and 5c(iii) were treated with $3 \times$ MIC and 5d(i), 5d(ii) and 5d(iii) were treated with $7 \times$ MIC concentration of JGP1 bioactive compound.

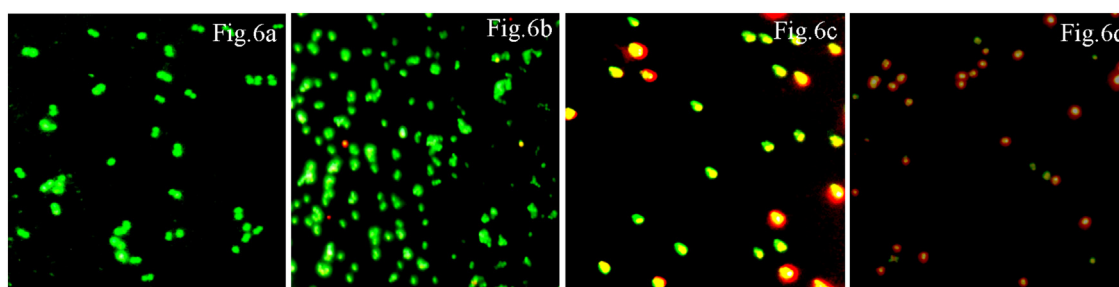


Fig. 6. MRSA cells stained with AO/PI imaged by fluorescent microscopy. (6a) Untreated, (6b) $1 \times$ MIC, (6c), $3 \times$ MIC and (6d), $7 \times$ MIC concentration of JGP1 respectively.

Interestingly, the isolate JRG-02 showed broad spectrum antibacterial characteristics against various pathogens. The effect of carbon source on the isolate JRG-02 revealed that the medium containing maltose may influence the biomass and antibacterial compound production when compared to lactose, a fairly significant component for the growth and antibacterial activity than other carbon sources. It has been reported that maltose as best carbon source for biomass and antibiotic production in case of *Streptomyces albidoflavus* [34]. Similarly, maximum biomass and antibacterial activity were obtained by *Streptomyces* sp. JRG-04 with maltose supplemented production medium [21]. The effect of nitrogen source on growth and antibacterial compound production of *Streptomyces* sp. JRG-02 was found to be tryptone. A similar result was observed

in the avilamycin production by *Streptomyces viridochromogenes* Tu 57-1 [35].

The physiological factors such as pH and temperature may influence the secondary metabolites production in several microorganisms. The production medium pH is essential for growth and significant product formation in the secondary metabolite producers [36]. Marine-derived *Streptomyces* sp. JRG-02 produces maximum antibacterial compound production at pH 7.5. Likewise, optimum pH for antimicrobial substance production of *Streptomyces* sp., KEH23 was observed at pH 7.5 [37]. The neutral pH has potential role for the growth and bioactive compound production of the strain. The temperature optimization results suggested that the optimum temperature for growth and antibacterial substance production for this strain was 28°C . In addition, the optimization

of sodium chloride concentration results revealed that, there is no further significant change in terms of antibacterial activity of the isolate JRG-02. Thus, the isolate grown well in presence of sodium chloride up to (w/v) 7.0%, however which fails to produce antibacterial compounds while increasing salt concentration above 1.5% and it's due to the unique physiological features of the isolate JRG-02. Similarly, the halophilic actinomycetes *Kocuria* sp. strain rsk4 grew well at the range of 5–10% (w/v) NaCl and which produces antibiotic optimally at 5% NaCl [38].

The antibacterial activity assay in terms of MIC of the active compound JGP1 revealed the efficiency of broad-spectrum activity against test pathogens. Remarkably, the active compound JGP1 had a significant activity against MRSA, a clinical pathogen. Live-dead analysis showed that, the antibacterial compound treatment on MRSA cells, may have growth inhibition, changes their membrane integrity followed by cell death. This result was supported using fluorescence microscopy and flow cytometry (FACS) analysis. Flow cytometer technique provides rapid evaluation of physiological status of bacterial pathogens when exposing antimicrobial compounds [39]. The present investigation using FACS results indicated the lethal activity of polyketide compound above MIC levels against MRSA cells showed noteworthy killing as compared to the control and MIC level.

This result is great agreement with previous reports [24,40]. In the FT-IR spectrum the absorption peaks at 3284 cm^{-1} , 1639.49 cm^{-1} and 1658.78 cm^{-1} indicated the presence of hydroxyl and ketone groups and this type of functional groups were important in aromatic polyketide. Hence, it might be the evident that the active compound JGP1 structurally related to polyketide antibiotic compound [7,41]. The ESI-MS results for the purified active compound of *Streptomyces* sp. JRG-02 had the mass value of 371 m/z .

Conclusions

The present study clearly shows that the survival of diverse inhabitants of *Streptomyces* in the mangrove sediments with potential biological activity and genetic diversity. *Streptomyces* sp. JRG-02 is a potent antibiotic candidate and chemically distinct entity with significant antibacterial activity against MRSA. The bioactive compound had the potential to breach the MRSA cell membrane. Fluorescence microscopic images and FACS analysis envisaged that PI uptake by the cells could be possible as the PI could bind to the DNA of membrane compromised cells. The further structural characterization of these molecules promotes the utilization of this active principle as a lead in case of drug discovery from marine based formulations to eradicate the emerging infectious drug-resistant pathogenic microorganisms.

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Competing interests

None declared.

Ethical approval

Not required.

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